

**Astrophysics and Cosmology/Astronomy and Astrophysics**  
(PH60046/PH40235)

Department of Physics, IIT Kharagpur  
**Mid Semester Examination, Spring 2024**

Date : 21<sup>st</sup> February, 2024 Time : 9.00-11.00 hrs

Answer all questions. No queries during exam. Write all issues in your answer script.

**Full Marks : 30**

1. Consider a main sequence star with parallax 0.02 arcsec has an apparent magnitude  $m = 4$  and radius  $R = 2.1 \times R_{\odot}$ .

- (a) What is the distance of the star from an observer on earth in pc?
- (b) Find out the luminosity of the star in units of  $L_{\odot}$ .
- (c) Determine the mass of the star in units of  $M_{\odot}$  using the Eddington luminosity relation.
- (d) If solar surface temperature is 5780 K, determine the surface temperature of the star in K?
- (e) Suppose the observed star is in a star-cluster and sits right below the main sequence turn off point in the H-R diagram. What is the age of the cluster in Myr? (Given the age of the sun is 4.5 Gyr)

[5×2 = 10]

2. Consider a star with radius  $R$  and density  $\rho_0$ .

- (a) Compute the mass of the star within radius  $r$ ,  $M(r)$ .
- (b) Determine the radial variation of the gravitational acceleration of the star,  $g(r)$ .
- (c) Write down the equation governing the hydrostatic equilibrium within the star.
- (d) If the pressure law of the stellar constituents follow the relation  $P = \kappa T^3$ . Determine the core temperature of the star.

[2+2+3+3 = 10]

3. Consider a star that is made up of adiabatic gas that obeys pressure and density relation  $P \propto \rho^{\gamma}$ , where  $\gamma$  is the adiabatic index. If  $H$  is the pressure scale height, use the ideal gas law and hydrostatic equilibrium to show that  $\left| \frac{dT}{dr} \right|_{ad} = \left( 1 - \frac{1}{\gamma} \right) \frac{T}{H}$ .

[5]

4. Draw the Hertzsprung–Russell diagram with appropriate axis and units. Identify the location of Main Sequence Stars, Giants Stars, Supergiants Stars, Horizontal Branch Stars and White Dwarfs.

[5]